MIL-F-51095B
30 SEPTEMBER 1964
SUPERSEDING
MIL-F-51095A(MU)
15 JULY 1963

MILITARY SPECIFICATION

FILTER MATERIAL, GAS-AEROSOL, C18

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of gas-aerosol filter material.

1.2 Classification. The filter material shall be of the following types as specified (see 6.2):

Type II — Low weight.

Type III — High weight.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636 — Box, Fiberboard.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-B-131 — Barrier Material; Water Vaporproof, Flexible.

MIL-F-50078 — Filter Element, Protective Mask, M13.

MIL-C-51096 — Core Material, C18.

MIL-B-51097 — Backing Material, C18.

STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

MIL-STD-414 — Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invi-

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tation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D–645 — *Thickness of Paper and Paper Products.*

D–828 — Tensile Breaking Strength of Paper and Paperboard.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Manufacture. The filter material shall be a uniform laminated sheet (see 6.2) consisting of several layers of core material conforming to MIL–C–51096, between single outside layers of backing material conforming to MIL–B–51097.

3.1.1 *Basis weight.* Prior to lamination, the bone dry basis weight (pounds per 3000 square feet) of the combined core layers shall be as follows using the basis weight designated on the outside wrap of the core material roll.

Type II — 645 pounds (minimum).
Type III — 1260 to 1380 pounds.

3.2 Physical and chemical requirements. The filter material shall conform to table I when tested as specified therein.

3.3 Delamination. Each disc taken as specified in 4.7.1 shall be an integral unit when died out from the test strip (see table II) and tested as specified in 4.7.4.7.

3.4 Preproduction sample. Prior to the start of regular production and following any change in regular production methods, materials, or equipment, sufficient filter material shall be manufactured using the same methods, materials and equipment as will be used during regular production, to produce 196 sheets (see 6.2) and 196 test strips. Test strips shall be no less than 10 inches long and shall be taken at the shears from the continuous run and no less than 3 feet apart. The preproduction sample shall pass the inspection specified in 4.3.

TABLE I. *Physical and chemical requirements*

Property	Requirement				Test method
	Type II		Type III		
	min	max	min	max	
Airflow resistance, mm water		35		50	4.7.2
DOP smoke pennetration, %		0.1		0.10	4.7.2
Accelerated gas sorption, min					4.7.3
Cyanogen chloride (CK)	16		12.5		
Chloropicrin (PS)	20		12.5		
Thickness, inches	0.17	0.19	0.30	0.33	4.7.4.1
Tensile strength, lbs/inch					4.7.4.2
Cross machine direction	7.0		12		
Moisture content, %		2.0		** 2.0	4.7.4.3
Flexibility, inches		0.10			4.7.4.4
* Diffusivity, cm/sec	.065				4.7.4.5
Flexure at –40° ± 5°F.	No cracks or breaks				4.7.4.6

* Not applicable to filter material for the CDV-805 civilian mask.

** A maximum moisture content of 5.0% is permitted for filter material fabricated into filter elements within six months of of date of manufacture of core material.

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3.5 Workmanship. The filter material shall be free from foreign matter, wrinkles, burn holes, scorching, charring, tears, cracks, holes, thin spots, blisters or misalignments when examined in accordance with 4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier's responsibility.* Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to specified requirements.

4.1.2 *Government's responsibility.* The Government will be responsible for the examination and tests in 4.3.1 and 4.3.2. Samples shall be forwarded to the laboratory designated by the contracting officer (see 6.2 and 6.7).

4.2 Special provisions.

4.2.1 ***Alternative inspection.*** **The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern (see 6.7).**

4.2.2 ***Objective evidence.*** **The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.**

4.2.3 ***Inspection substation.*** **If the supplier elects to establish inspection substations for the purpose of conducting inspection for certain listed defects, the Acceptable Quality Levels (AQL's) for each inspection substation shall be equal to the AQL which is the next numerically lower value than the AQL specified for the total listed defects. The requirements of MIL–STD–105 are applicable to each substation and to the end item.**

4.3 Preproduction sample inspection.

4.3.1 ***Nondestructive examination.*** **Fifty sheets shall be taken at random and examined in accordance with 5.6.2.1.**

4.3.2 ***Destructive tests.*** **Test strips shall be taken at random and tested as specified in table II.**

4.3.3 ***Acceptance/rejection criteria.*** **The preproduction sample shall meet all the requirements of 4.3.1 and 4.3.2 to be acceptable.**

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TABLE II. *Tests*

Property	Test strips	Test specimens (see 4.7.1)	Retest	Defective specimen	Action	Test and retest combined: Failures	Test and retest combined: Action	Test method
Preproduction sample				5	Accept			
Airflow resistance	150	150	0	6	Reject			4.7.2
DOP smoke penetration	150	150	0	5 6	Accept Reject			4.7.2
Accelerated gas sorption		CK 34 PS 34	0		(*)			4.7.3
Physical properties	20	10 each property	10 each property	0–2 3–5 6	Accept Retest Reject	3″5 6	Accept Reject	4.7.4
Delamination	150	150	0	5 6	Accept Reject			4.7.4.7
Regular production:								
Airflow resistance	AQL 1.5 percent defective MIL–STD–105							4.7.2
DOP smoke penetration	AQL 1.5 percent defective MIL–STD–105							4.7.2
** Accelerated gas sorption CK PS	(*)	1 per strip 1 per strip			(*)	See 4.8		4.7.3
Physical properties	20	10 each property	10 each property	0–2 3–5 6	Accept Retest Reject	3–5 6	Accept Reject	4.7.4
Delamination	AQL 1.5 percent defective MIL–STD–105							4.7.4.7

* Use MIL-STD 414 with Inspection Level II: Sampling Plan for Single Specification Limit-Form 1, with Variability Unknown-Standard Deviation Method; and an AQL of 1.50 percent; Unit of product-one sheet; for preproduction sample use K factor of 1.76. In no case shall the sample size be less than the size indicated for code letter H in MIL-STD-414.

** Test specimens shall be taken at random from those specimens which have passed 4.7.2.

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4.4 **Lotting.** A lot shall consist of the filter material produced during one work shift not exceeding 12 hours by one manufacturer under essentially the same manufacturing conditions. No more than two lots of core material produced by one manufacturer shall be used in one lot of filter material.

4.5 **Sampling.**

4.5.1 *For examination.* Sampling shall be conducted in accordance with MIL–STD–105.

4.5.2 *For destructive tests.* Test strips (minimum 10-inch length taken at the shears from the continuous run and no less than 3 feet apart) shall be taken at random for testing as specified in table II.

4.6 **Inspection provisions.**

4.6.1 *For examination.* Sample sheets of filter material and the level A and level B preparation for delivery shall be examined in accordance with the classification of defects and MIL–STD–105.

4.6.2 *Classification of defects.*

4.6.2.1 *Filter material, gas-aerosol.*

Categories	*Defects*
Critical:	None defined.
Major:	AQL 0.65 percent defective.
101	Damage (tears, cracks, holes, blisters or thin spots)
102	Foreign matter
103	Burn holes, charring, or scorching
104	Wrinkles
Minor:	AQL 2.5 percent defective
201	Charcoal on top or bottom surfaces
203	Misalignment of core material exceeding ¼ inch

4.6.2.2 *Preparation for delivery (sec. 5).*

Categories	*Defects*
Critical:	None defined.
Major:	AQL 1.0 percent defective
101	Filter material not inclosed within heat-sealed barrier bags as specified
102	Water vaporproof barrier bags incorrect or damaged
103	Shipping container closure incorrect
104	Shipping container damaged (broken, crushed, punctured, or torn)
105	Marking incorrect, missing, or illegible
Minor:	AQL 2.5 percent defective
201	Shipping containers incorrect
202	Filler pads missing
203	Tape missing or incorrect

4.7 **Tests.**

4.7.1 *Preparation of test specimens.* One 5¼ inch diameter disc shall be taken from each test strip for airflow resistance and DOP tests. Discs and specimens for physical tests shall be taken at random locations from test strips but not within one inch of the edge. Specimens for testing moisture content shall be taken immediately prior to heat-sealing finished packs (see 5.1) and these specimens shall be sealed immediately in packages in the same manner as the finished packs.

4.7.2 *Airflow resistance and DOP smoke penetration.* Airflow resistance and DOP smoke penetration of the specimens shall be determined at a flow-rate of 32 liters per minute using the Edgewood Arsenal Q127 DOP Penetrometer.

4.7.3 *Accelerated gas sorption.* The accelerated gas sorption of each specimen shall be determined using Edgewood Arsenal Q95 Gas Life Testing Apparatus with the test conditions of table III. The accelerated gas sorption of each specimen shall be corrected for variations in concentration and reported to the nearest 0.1 minute.

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TABLE III. *Test conditions*

Filter material	Test gas	Concentrations *	Flow rate	Relative humidity for equilibrating and testing
Type II	CK	1 mg/l	12.5 l/min, Intermittent flow	80–80%
	PS	4 mg/l	12.5 l/min, Intermittent flow	50–50%
Type III	CK	4 mg/l	12.5 l/min, Intermittent flow	80–80%
	PS	50 mg/l	8 l/min, continuous flow	0–50%

* The indicated test gas concentration shall be maintained within ± 5%. Equilibrating and test temperatures will be recorded and maintained at 80° ± 10°F.

4.7.4 *Physical properties.*

4.7.4.1 *Thickness.* Three thickness readings shall be taken on each test disc (see 4.7.1) in accordance with Method C of ASTM D645 except that conditioning of specimens shall be omitted. The thickness of a specimen shall be the average of 3 readings.

4.7.4.2 *Tensile strength.* Tensile strength of the test specimens shall be determined in accordance with ASTM D828, except that conditioning of specimens shall be omitted.

4.7.4.3 *Moisture content.* Each test specimen (5¼-inch diameter disc) shall be weighed to the nearest 0.01 gram (wet weight). Place the specimens in a forced air drying oven at 105° to 110°C. and a maximum of 20 percent relative humidity until constant dry weight, within 0.01 gram, is reached. Calculate the percent moisture as follows:

$$\text{Percent moisture content} = \frac{100\ (\text{wet weight-dry weight})}{\text{wet weight}}$$

4.7.4.4 *Flexibility.* One specimen, 1 by 6 inches, shall be died at random locations from each of 10 of the sample sheets. Five shall be died in the machine direction and five in the cross machine direction.

4.7.4.4.1 *Apparatus.* Flexibility shall be determined by means of an Aminco "Modulimeter."

4.7.4.4.2 *Procedure.* Level the test apparatus, adjust its anvils to a span of 4 inches and determine the deflection, in thousandths of an inch, imparted to each test specimen by a 0.10 pound load applied midway between the anvils by a presser foot 1.375 inches long, and 0.25 inches diameter.

4.7.4.5 *Carbon dioxide diffusivity.* One specimen, 13½ by 13½ inches, shall be cut at random locations from each of 10 selected sheets.

4.7.4.5.1 *Apparatus.* Use a diffusion chamber conforming to figure 1 and a Holdane-Guthrie gas analysis apparatus.

4.7.4.5.2 *Procedure.* Perform the test as follows:

(a) Connect the gas analysis apparatus to the capillary tube of the diffusion chamber.

(b) Seal the specimen, without distortion, to the bottom of the chamber; then block off the outside area of the specimen, using an impervious barrier material placed against the outer surface and held by suitable means.

(c) Pass into the chamber sufficient carbon dioxide, at a temperature of 75° ±3°F., to yield a concentration greater than 10 percent.

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(d) After an interval of 5 minutes, remove the outside barrier and measure the carbon dioxide concentration in the chamber.

(e) Continue to measure the carbon dioxide concentration in the chamber, at intervals of 3 minutes, until it is below 2 percent.

(f) Plot concentration versus time on semi-log paper and construct the best straight line of the plotted values covering the range between 2 and 8 percent carbon dioxide.

(g) Calculate the diffusivity as follows:

$$\text{Diffusivity} = 1.386 \frac{V}{AT}$$

where: V = volume of chamber (cu. cm.)

A = area of exposed specimens (sq. cm.)

T = elapsed time (seconds) required for the carbon dioxide concentration to drop from 8 to 2 percent according to the straight line graph.

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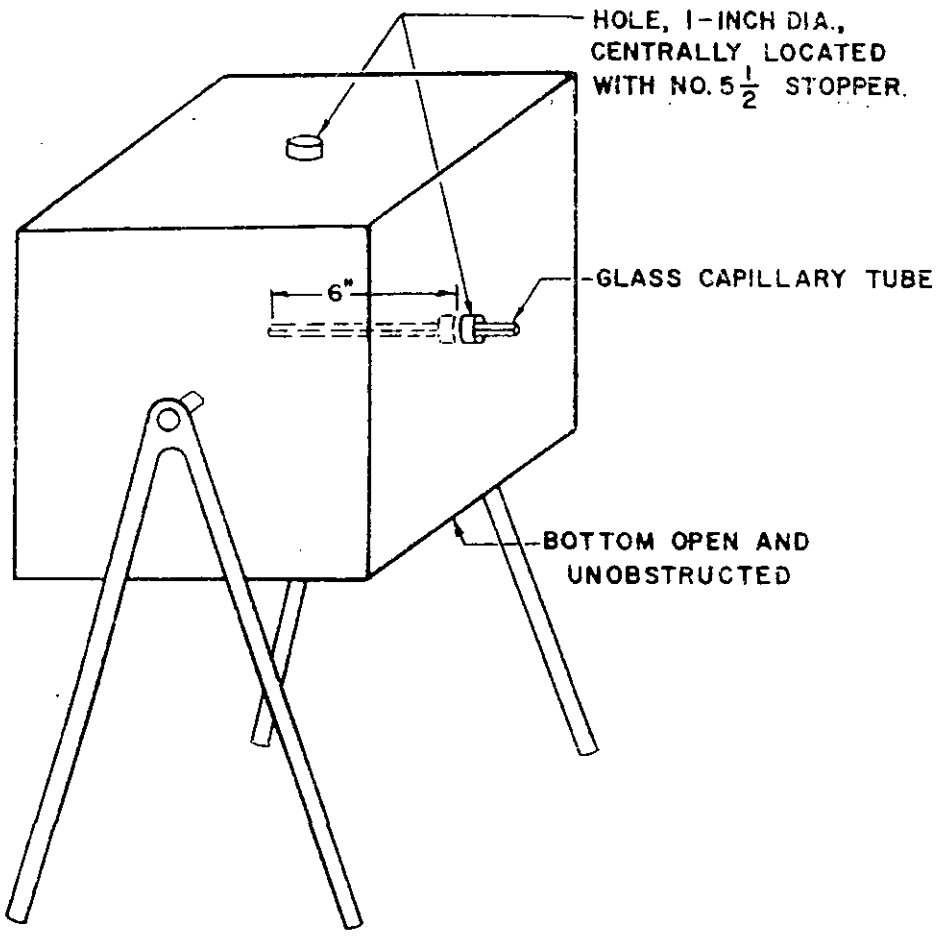


Figure 1- DIFFUSION CHAMBER

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4.7.4.6 *Flexure.* One specimen, 5 by 5 inches, shall be died at random locations from each of 10 of the sample sheets. Five shall be flexed in the machine direction and five in the cross machine direction.

4.7.4.6.1 *Procedure.* The specimens shall be equilibrated at —40° ±5°F. and each specimen shall be flexed once through 180° (folded) at —40° ± 5°F.

4.7.4.7 *Delamination.* The test specimen (5¼ inch diameter disc) shall be lifted from a flat surface so that the lower backing layer is not held by the fingers. The test shall be repeated on the inverted disc. The test specimen shall be considered an integral unit if neither the top nor bottom backing layer separates completely from the core material.

4.8 **Acceptance/rejection criteria (Type III only).** If the quantity $\bar{x}$—ks, 11 minutes ($\bar{x}$, k, s obtained from MIL–STD–414), then 20 matched pairs of filter elements shall be manufactured in accordance with MIL–F–50078 from sample sheets of the lot of filter material in question. These elements shall be tested for chloropicrin gas life before and after rough handling as specified for regular production in MIL–F–50078. The lot of filter material shall be accepted if each group of 10 pairs of elements meets the chloropicrin gas life requirements specified in MIL–F–50078.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**

5.1.1 *Level A.* A uniform quantity of filter material sheets of a single type shall be packed, not to exceed a gross weight of 40 pounds, in a W5c, style FTC, fiberboard box, conforming to PPP–B–636. The entire quantity of filter material that is to be packed within one box shall be inclosed within two heat-sealed water vaporproof barrier bags with a filler pad fabricated from the same material as the box, placed on the bottom and top of each stack of filter material sheets placed inside the barrier bags. The first bag shall be in intimate contact with the filter material and filler pads. The second bag shall be in intimate contact with the first bag. The barrier bags shall be fabricated from class 1, water vaporproof barrier-material, conforming to MIL–B–131. Closure and test requirements for these barrier bags shall be the same as specified for method IA–14 of MIL–P–116. The fiberboard boxes with the bagged filter material packed therein shall then be closed and reinforced in accordance with the appendix requirements of PPP–B–636 for exterior boxes. Tape banding shall be used in lieu of metal banding as the reinforcement requirement (see 6.3).

5.1.2 *Level B.* Level B packing shall be identical with level A except that the filter material shall be inclosed in a single heat-sealed water vaporproof bag instead of the double bag (see 6.4).

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with MIL–STD–129 (see 6.5). All containers shall be marked as follows:

STORE IN A COOL, DRY LOCATION

6. NOTES

6.1 **Intended use.** Type II filter material is intended for use in the CDV–805 Civilian Protective Mask; CDV–815 Infant Protector; and the M18 Headwound Protective Mask. Type III filter material is intended for use in the ABC–M17 Field Protective Mask.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required.

(c) Type and intended use of filter material.

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(d) Sheet size of material.

(e) The laboratory to which samples are to be sent.

6.3 **Moisture content.** Filter material should be packed immediately as it comes off the production line in order to prevent moisture pick-up.

6.4 **Packing, level B.** Level B packing is intended to provide economical but limited protection, and should be specified only when it is determined that the filter material will be held in a cool location for a period not to exceed six months from the date of initial packing.

6.5 **Marking of filter material.** Ink or other marking fluid containing organic solvents should not be used on the filter material as organic solvents penetrate and affect the adsorptive capacity of the material.

6.6 **Manufacturing tests.** The Government will fabricate filter elements from the preproduction sample of type III filter material in accordance with MIL-F-50078. These elements will be tested for "Plastisol edgeseal," "DOP smoke penetration," "Airflow resistance," and "Gas life" requirements of MIL-F-50078.

6.7 **Nonmilitary procurement.** For nonmilitary procurement, the procuring agency shall be responsible for preproduction sample inspection and for approval of alternative inspection procedures.

6.8 **Significant places.** For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM *Designation*: E29).

6.9 **International interest.** Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodians:
Army—MU
Navy—SH
Air Force—84

Preparing activity:
Army—MU(EA)
Project No. 4240-0111

International interest: (See 6.9)

Review activities:
Army—MD
Navy—SH

User activities:
Navy—MC
Air Force—84

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